

DO NOT WRITE ON TEST



**FINANCIAL MATH &
ANALYSIS CONCEPTS**

~OPEN EVENT~

(190)

STATE 2023

TOTAL POINTS

_____100 Points

Test Time: 60 minutes

GENERAL GUIDELINES:

Failure to adhere to any of the following rules will result in disqualification:

1. Member must hand in this test booklet and all printouts if any. Failure to do so will result in disqualification.
2. No equipment, supplies, or materials other than those specified for this event are allowed in the testing area. No previous BPA tests and/or sample tests (handwritten, photocopied, or keyed) are allowed in the testing area.
3. Electronic devices will be monitored according to ACT standards.
4. For any interest calculations, assume a 360-day year.
5. Unless otherwise stated, assume that all plant assets were purchased on January 1.
6. Unless otherwise stated, assume that the overtime rate is 1.5 regular time pay for any hours worked in excess of 40 hours per week.

Multiple Choice Questions

Directions: Identify the letter of the choice that best completes the statement or answers the question.

1. What amount of tax will be owed on an income of \$63,280 if the city income tax rate is 1.5% of all income?
 - A. \$94.92
 - B. \$245.40
 - C. \$465.12
 - D. \$949.20

2. What are the gross earnings of an employee who works 46 hours and makes \$11.23 per hour? Overtime is paid for hours over 40.
 - A. \$101.70
 - B. \$449.20
 - C. \$516.58
 - D. \$550.30

3. Nell borrows \$4,500 at 6% for 24 days. What amount of interest does Nell pay?
 - A. \$270
 - B. \$27
 - C. \$18
 - D. \$12

4. The property tax rate is \$34.67 per \$1,000. Find the tax due on a business whose property is assessed at \$540,000.
 - A. \$18,721.80
 - B. \$1,872.18
 - C. \$1,215.80
 - D. \$187.22

5. On Tuesday, Henry used a debit card to pay for garden tools, \$83.12; work clothing, \$46.75; groceries, \$54.79. If Henry's bank balance was \$437.01 at the start of the day, what is his new balance at the end of the day?
 - A. \$197.62
 - B. \$252.35
 - C. \$307.14
 - D. \$621.67

6. The Jensen family has a total yearly budget of \$48,000. If the Jensen family budgets \$4,500 for medical costs, what percent of their total budget did the budget for medical costs, to the nearest percent?
- A. 7%
 - B. 8%
 - C. 9%
 - D. 10%
7. Carmen is paid piece-rate for each of the 268 items she produces in a week, and she received gross wages of \$469. What is Carmen's per piece rate?
- A. \$1.74
 - B. \$1.75
 - C. \$2.24
 - D. \$2.25
8. Jontay works 8 hours a day Monday through Friday. What is his gross pay for the week if he makes \$9.84 per hour?
- A. \$314.88
 - B. \$344.40
 - C. \$383.60
 - D. \$393.60
9. Olga Kirosky deposited these items in a bank: (bills) 12 twenties, (coins) 11 quarters, 16 dimes; (checks) \$125.99, \$41.65. She received 100 one-dollar bills in cash back. What was the net amount of her deposit?
- A. \$311.99
 - B. \$319.99
 - C. \$411.99
 - D. \$419.99
10. Juan bought three pairs of gloves at \$15.89 each and two pair of shoes at \$78.98 each. There was a state tax of 4.75%. What was the total of the cash register receipt?
- A. \$205.63
 - B. \$215.40
 - C. \$234.35
 - D. \$252.98
11. Carl found an automatic washer priced at \$549 and a dryer for \$399. They are on sale for only \$849 for the set. How much can Carl save if he buys the items as a set?
- A. \$69
 - B. \$79
 - C. \$89
 - D. \$99

12. The mean is the sum of the numbers divided by the number of items.
- A. True
 - B. False
13. Vase Inc. finds that 14 out of 200 glass vases inspected are defective. Assume there are no changes in the way vases are made. How many defective vases could be expected from a month's usual production of 7,000 glass vases?
- A. 310
 - B. 390
 - C. 490
 - D. 510
14. During the last 365 days, a weather forecaster's predictions have been right 292 days, leaving 73 days with the wrong forecast. What is the probability that the forecaster's prediction of tomorrow's weather will be right?
- A. 20%
 - B. 55%
 - C. 70%
 - D. 80%
15. Alexis was injured on her job. She is insured by a disability policy that pays 35% of her average annual salary of \$36,800. What amount does Alexis collect monthly?
- A. \$975.68
 - B. \$1,073.33
 - C. \$2,142.83
 - D. \$12,880.00
16. Visual Graphics, Inc. can buy a spreadsheet program for \$89.99 per license. It can also buy a shareware spreadsheet program for \$39.99 per license with fewer features. If they buy the shareware for their 21 computer systems, how much will they save?
- A. \$50.00
 - B. \$105.00
 - C. \$1,050.00
 - D. \$1,500.00
17. Mia worked 8 hours per day, Monday through Wednesday, and 10 hours per day on Thursday and Friday. If she is paid \$13.50 regularly, what was her gross pay for the week?
- A. \$81
 - B. \$540
 - C. \$594
 - D. \$621

18. TriStar, Inc. leased 15 computer systems for three years. The monthly lease for each was \$48.99. What was the total cost of leasing all the notebooks for three years?
 - A. \$26,454.60
 - B. \$15,462.64
 - C. \$8,818.20
 - D. \$1,763.64

19. Copy paper costs \$4.15 per ream (500 sheets). When bought in a case of 10 reams, the same paper costs \$38 per case. What is the savings per ream when buying a case?
 - A. \$.28
 - B. \$.35
 - C. \$1.23
 - D. \$3.50

20. One office in an office building has an area of 5200 ft². The office costs are \$87,360 a year including rent, heating, cooling, lighting, and maintenance. What is the annual cost of the office per square foot?
 - A. \$16.80
 - B. \$168.00
 - C. \$1,680.00
 - D. \$11,680.00

21. Clarence put 1,489 miles on his car last month. Of those miles, 27% were for his job. The company reimburses Clarence \$.35 a mile for business use. What did Clarence receive as reimbursement for the use of his car last month?
 - A. \$14.07
 - B. \$140.71
 - C. \$224.46
 - D. \$521.15

22. Cost-of-Living Adjustment (COLA) is a wage decrease based on changes in the Consumer Price Index (CPI).
 - A. True
 - B. False

23. Amari went out to eat with his friends. He decided to pay for the everyone's meal and leave a 22% tip. Amari's meal was \$24.63. Janet's meal was \$19.55. Tanner's meal was \$32.56. What was the total cost that Amari spent for all the meals and tip?
 - A. \$16.88
 - B. \$76.74
 - C. \$93.62
 - D. \$96.72

24. Farwell Painting Company received a 2/10; n/30 invoice that they paid on day 8. The total on the invoice was \$2,460. What did Farwell pay on this invoice?
- A. \$246.00
 - B. \$2,214.00
 - C. \$2,410.80
 - D. \$2,460.00
25. Truong's favorite clothing store was running a BOGO 50% off sale on everything. You must pay the full price of the highest ticket item and the second will be discounted 50%. He bought a pair of jeans for \$119.25 and a jacket for \$89.20. There is a 5.4% sales tax on all goods. What did he pay for his jeans and jacket?
- A. \$172.70
 - B. \$163.85
 - C. \$154.97
 - D. \$126.50
26. A retailer bases its markup of shoes on cost . What is the selling price of a shoe with a cost of \$56 and a 30% markup?
- A. \$59.50
 - B. \$67.00
 - C. \$72.80
 - D. \$80.40
27. A business had net sales of \$140,000. The cost of goods sold was \$84,000. What percent of the net sales was the cost of goods sold?
- A. 40%
 - B. 50%
 - C. 60%
 - D. 70%
28. Juliette earned \$274.50 last week at her regular job. She also worked 11 hours at her part-time job that pays her \$9.25 an hour. What total pay did she earn for both jobs?
- A. \$283.75
 - B. \$295.45
 - C. \$346.35
 - D. \$376.25
29. Asia found that her job expenses for last year were: uniforms \$329; licenses, \$278; professional dues, \$475; commuting costs, \$1077. Her total job benefits for the same period were \$56,102. What is Asia's net job benefits?
- A. \$58,261
 - B. \$53,943
 - C. \$21,590
 - D. \$2,159

30. Rosa borrows \$1,000 at 6% interest for 85 days. How much will she spend in interest?
- A. \$6.00
 - B. \$8.50
 - C. \$11.45
 - D. \$14.17
31. A repairperson installed eight sections of cedar fencing for Jose Velez's back yard. Jose was charged \$462.53 for the sections and \$225 for the installation. A sales tax rate of 6.75% was charged on goods but not labor. How much was his total bill for the fence?
- A. \$687.53
 - B. \$702.72
 - C. \$718.75
 - D. \$733.94
32. Tito becomes completely disabled. His group disability policy's benefit percentage is 60% of his average annual salary for the last 3 years. His annual salaries were \$35,800, \$38,500, and \$39,800. What annual benefit amount will he receive?
- A. \$22,820
 - B. \$25,680
 - C. \$33,450
 - D. \$38,033
33. Atom Up Company placed an ad for control employees for its new plant. The cost of designing the ad was a one-time charge of \$230. The ad appeared in a local newspaper for two days during the week and again on Sunday. The newspaper's charge for the ad was \$680 a day during the week and \$810 on the weekend. What amount did Atom Up Company spend for advertising?
- A. \$1,360
 - B. \$1,720
 - C. \$2,400
 - D. \$2,640
34. Jeron worked the following schedule:
- | | |
|-----------|-----|
| Monday | 9-4 |
| Tuesday | 9-3 |
| Wednesday | 9-5 |
| Thursday | 9-2 |
| Friday | 9-6 |
- Jeron gets paid lunch break every day he works. He makes \$14.95 an hour. What did he make last week in gross pay?
- A. \$508.30
 - B. \$523.25
 - C. \$538.20
 - D. \$553.15

35. The part of a price that is paid at the time of buying on the installment plan is called the _____.
- A. principle
 - B. interest
 - C. exemption
 - D. down payment
36. Shaina bought these items at an office supply store: 2 dozen pencils at \$1.99 each, 8 boxes of cellophane tape at \$5.99 each, 3 cartons of envelopes at \$9.83 each. The store applied a 7.2% sales tax. What was the total of the sales slip?
- A. \$79.50
 - B. \$81.39
 - C. \$87.25
 - D. \$92.57
37. Adele earns a salary plus commission. Her total earnings last year were \$47,600, of which \$16,800 was salary. Her total sales last year were \$616,000. What was her rate of commission?
- A. 5%
 - B. 6%
 - C. 7%
 - D. 8%
38. A 50-oz. bottle of liquid detergent sales for \$3.49. The unit price per ounce is \$.07.
- A. True
 - B. False
39. TeleSpeed is an internet service provider that offers Yang the internet for \$12 a month. Yang also has to pay a \$15 set-up fee. Yang pays an additional \$29.95 a year to another firm for antivirus software. What will be Yang's total cost of connecting for one year?
- A. \$144
 - B. \$159
 - C. \$173.95
 - D. \$188.95
40. Winston borrows \$3,000 at 7% interest for 43 days. What interest will he pay on the loan?
- A. \$12.25
 - B. \$21.00
 - C. \$25.08
 - D. \$30.00

41. Lela's employer offers a health insurance plan that covers Lela, her husband and their child. The total monthly premium is \$285, of which the employer pays 26%. How much does Lela pay for the health insurance for one year?
- A. \$2,245.60
 - B. \$2,530.80
 - C. \$3,420.00
 - D. \$3,945.60
42. Find the median for this set of numbers: 9, 0, 1, 9, 5, 4, 9, 2.
- A. 4
 - B. 4.5
 - C. 5
 - D. 5.5
43. Angelica has gross pay of \$617.50 a week. How much will she earn in one quarter of a year at her current pay?
- A. \$2,470.00
 - B. \$7,410.00
 - C. \$8,027.50
 - D. \$10,246.50
44. The tip income of the seven food servers at a buffet restaurant averages 7% of the food bill. The total of the food bills for one evening was \$8,200. How much did each food server earn for the evening?
- A. \$82
 - B. \$87
 - C. \$124
 - D. \$574
45. Quentin is paid \$1.50 for each unit he assembles. During the week, he assembled these units: 64, 68, 59, 62, and 66. What was his gross pay for the week?
- A. \$278.50
 - B. \$319.00
 - C. \$405.50
 - D. \$478.50
46. A newspaper charges \$67.20 a line per day for employment advertising. If Company XYZ placed a 13-line ad for two days, how much would they spend?
- A. \$134.40
 - B. \$873.60
 - C. \$1,747.20
 - D. \$2,241.80

47. Inez agrees to a purchase price of \$23,050 for a new car. The sales tax is 3.5% of the purchase price and registration fees are \$118. Inez made a 20% down payment of the purchase price. What is the delivered price of the car?
- A. \$15,475.20
 - B. \$17,340.40
 - C. \$17,564.75
 - D. \$23,974.75
48. The Maximus Corporation decides to give a bonus to Lowell Jacobs. The bonus is 3.9% of his annual earnings of \$64,580. What amount of bonus will Lowell receive?
- A. \$2,515.50
 - B. \$2,518.62
 - C. \$3,215.50
 - D. \$3,657.62
49. The insurance usually obtained by tenants to cover the things they own and to provide personal liability protection is known as a(n) _____.
- A. renter's insurance
 - B. homeowners' insurance
 - C. property damage insurance
 - D. comprehensive insurance
50. After the holidays, you can buy boxes of greeting cards at 2 for \$32. The same boxes sold during the season for \$23.99 each. If you bought 6 boxes after the holidays, how much would you save?
- A. \$143.94
 - B. \$96.00
 - C. \$47.94
 - D. \$23.99